

LESSON-19

TRIGONOMETRIC EQUATION

- $\cos n\pi = -1$ ($n = \text{odd}$)
 $= 1$ ($n = \text{even}$)

- $\sin n\pi = 0$ ($n = \text{Any Integer}$)

$$\sin \frac{n\pi}{2} = 1$$

$$\cos \frac{n\pi}{2} = -1 \quad (\text{Even, Multiple of } 2, 6, 10, \dots)$$

$$= +1 \quad (\text{Even, Multiple of } 4, 8, 12, \dots)$$

Sin cos Tan $\longrightarrow$ General Solution &

i. $\sin x = \sin \alpha$

$$\boxed{x = n\pi + (-1)^n \alpha} \quad \text{--- (i)}$$

ii. $\cos x = \cos \alpha$

$$\boxed{x = 2n\pi \pm \alpha} \quad \text{--- (ii)}$$

iii. $\tan x = \tan \alpha$

$$\boxed{x = n\pi + \alpha} \quad \text{--- (iii)}$$
